OceanVerse AI: AI Driven Unified Oceanographic & Biodiversity Platform

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Abstract— OceanVerse AI is a unified prototype platform designed to integrate oceanographic monitoring, marine biodiversity information, environmental analytics, satellite observations, Internet of Things (IoT) sensing, artificial intelligence (AI), interactive mapping, and safety alerts in a single system. Existing marine information systems often address individual tasks or depend on heterogeneous data sources, making integrated analysis difficult. The proposed platform organizes key ocean parameters such as temperature, salinity, turbidity, pH, and chlorophyll and provides a modular dashboard for monitoring and analysis. A Random Forest classifier is implemented in the current prototype for demonstration-level algal bloom prediction using temperature, salinity, turbidity, chlorophyll, and pH as input features. Sentinel-2 Level-2A satellite imagery is also processed using the Normalized Difference Water Index (NDWI) for water-area detection. On the demonstration dataset, the implemented Random Forest model obtained 100% accuracy, precision, recall, and F1-score on the held-out test set. These values are specific to the demonstration dataset and should not be interpreted as real-world deployment performance. The prototype provides a foundation for larger real-time datasets, sensor streams, advanced AI models, and cloud services.

Index Terms— Artificial Intelligence, Ocean Monitoring, Marine Biodiversity, Random Forest, Sentinel-2, NDWI, IoT, Ocean Safety

I. INTRODUCTION

Ocean ecosystems are affected by pollution, climate variability, biodiversity loss, extreme weather, and other environmental pressures. Useful marine information is distributed across observations, databases, satellite products, sensor networks, and research systems. OceanVerse AI addresses this fragmentation by providing a unified platform for monitoring, analysis, visualization, and decision support.

OceanVerse AI is intended for fishermen, boat operators, researchers, travelers, tourists, and coastal communities. The platform combines oceanographic monitoring, biodiversity analysis, environmental prediction, satellite analysis, IoT monitoring, interactive mapping, and alerts. It also supports climate awareness and sustainable ocean management.

II. RELATED WORK

Previous research demonstrates the value of structured marine information systems, biodiversity databases, remote sensing, machine learning, autonomous sensing, and AI-enabled marine monitoring. Ontology-based approaches support common semantics, knowledge sharing, and interoperability across domains [1]. JMarine demonstrated a web-based marine biodiversity database with species, taxonomy, spatial, bathymetry, and mapping information, but its scope was regional [2].

Marine biodiversity studies have emphasized standardized data, validation, sampling, and spatial-temporal analysis [3], [4]. Ocean connectivity research shows that oceanographic processes influence biodiversity patterns [7]. Standardized marine taxon image databases can support future AI-based image identification [8]. Global spatial analyses also show the value of combining heterogeneous datasets to study biodiversity pressures [9].

AI and automated sensing are increasingly used in marine monitoring. Computer-vision monitoring, remote sensing, machine learning, marine protected-area governance, ocean big-data systems, and AI-based biodiversity analysis demonstrate the potential of technology while also highlighting interoperability, data-quality, cost, and standardization challenges [10]–[17].

III. PROBLEM STATEMENT

Existing marine applications frequently focus on a specific problem, species, geographic region, or data source. Ocean observations may be distributed among separate platforms, while biodiversity, satellite, weather, and sensor information may use different formats and access methods. This fragmentation makes combined environmental analysis and decision support difficult.

The project therefore addresses the problem of integrating heterogeneous oceanographic, biodiversity, satellite, and sensor information into one accessible platform that also provides AI-assisted analysis and visual decision support.

IV. OBJECTIVES

1) Develop a unified web platform for oceanographic and biodiversity information.

2) Monitor temperature, salinity, turbidity, pH, and chlorophyll.

3) Implement a demonstration AI component for algal bloom classification.

4) Process Sentinel-2 Level-2A imagery and demonstrate water detection using NDWI.

5) Provide an interactive ocean map with conceptual Safe, Caution, and High-Risk zones.

6) Provide an architecture for future IoT streams, environmental data, alerts, and reports.

7) Support data-driven marine conservation and safer decision making.

V. SYSTEM ARCHITECTURE

OceanVerse AI follows a modular data-to-decision architecture. Data sources may include ocean observations, weather information, pollution and environmental datasets, Sentinel-2 satellite products, and IoT sensor readings. The backend uses Python and FastAPI, while PostgreSQL provides structured data storage. Data-processing and AI components operate between the data and presentation layers.

The project technology specification includes React.js and Tailwind CSS for the interface; FastAPI and Python for the backend; TensorFlow as part of the planned AI/ML stack; Pandas and NumPy for data processing; PostgreSQL for the database; Leaflet.js for maps; AWS for cloud deployment; Firebase for authentication; Docker for containerization; Chart.js/Recharts for visualization; and Git/GitHub for version control.

VI. MODULES

A. Unified Dashboard: summarizes ocean conditions and provides navigation to the major modules.

B. Oceanographic Monitoring: displays temperature, salinity, turbidity, pH, and chlorophyll.

C. Biodiversity Monitoring: organizes species, habitat, and ecosystem-health information.

D. AI Prediction and Analysis: the current prototype implements algal bloom classification using Random Forest. Other proposed predictions require additional datasets and trained models.

E. Satellite Analysis: processes Sentinel-2 Level-2A bands and uses B03 Green and B08 NIR for NDWI-based water detection.

F. IoT Monitoring: provides the interface for sensor readings such as temperature, salinity, and turbidity; physical ESP32/Arduino integration is a future extension.

G. Interactive Ocean Map: uses Leaflet.js to display monitoring locations and conceptual safety zones.

H. Alerts and Reports: presents environmental warnings, prediction messages, and summarized results.

VII. AI/ML METHODOLOGY

The current AI experiment uses a Random Forest Classifier. The input features are temperature, salinity, turbidity, chlorophyll, and pH, and the target variable is an algal-bloom class represented as 0 or 1. The data are divided into training and test sets, the classifier is trained, and standard classification metrics are calculated.

Random Forest is an ensemble classification method that combines multiple decision trees. It is suitable for tabular feature-based classification and provides a practical baseline for a prototype. The current result demonstrates the end-to-end AI pipeline rather than establishing a scientifically validated operational algal-bloom forecasting model.

The project specification lists TensorFlow as part of the broader AI/ML technology stack. The current implemented demonstration classifier is a Scikit-learn Random Forest model; this distinction should be maintained in technical reporting.

VIII. SATELLITE DATA PROCESSING

Sentinel-2 Level-2A products provide atmospherically corrected surface reflectance together with quality information such as scene classification layers. The 10 m product includes B02, B03, B04, and B08 [18].

For the prototype, B03 (Green) and B08 (Near Infrared) are used to compute the Normalized Difference Water Index:

NDWI = (Green - NIR) / (Green + NIR).

Positive NDWI values are used as a simple indicator of likely water pixels. NDWI is a remote-sensing index, not an AI model. Recent work has also explored combining NDWI with Random Forest for water-body detection [19].

For rigorous quantitative remote sensing, product metadata and band-specific radiometric offsets should be applied when converting digital numbers to surface reflectance. The current workflow is therefore presented as a prototype demonstration.

IX. DATABASE AND API DESIGN

PostgreSQL is used for structured ocean observations. A representative record contains location, temperature, salinity, turbidity, chlorophyll, pH, and recording time. The FastAPI backend exposes endpoints for ocean data, database records, and satellite-analysis results. This separation keeps the frontend independent from processing and storage layers.

X. RESULTS

The implemented Random Forest demonstration produced the following held-out test-set results:

TABLE I. AI CLASSIFICATION RESULTS

Metric — Result

Accuracy — 100%

Precision — 100%

Recall — 100%

F1-Score — 100%

Model — Random Forest Classifier

XI. DISCUSSION

The 100% values were obtained on the held-out test set of the demonstration dataset. Because the dataset is limited and designed for a prototype, these values should not be generalized to real-world ocean conditions. Real deployment would require large, geographically diverse, temporally representative, independently validated observations.

The satellite workflow demonstrates Sentinel-2 band extraction and NDWI calculation for water detection. The interactive map provides a practical spatial interface, while the modular architecture allows future integration of live sensor and environmental feeds.

XII. LIMITATIONS

1) The AI model is a demonstration model trained on a limited prototype dataset.

2) The satellite module demonstrates NDWI water detection rather than a complete AI satellite-classification system.

3) IoT hardware integration requires physical sensors and communication infrastructure for live deployment.

4) Proposed functions such as cyclone-impact forecasting, fish-availability prediction, and advanced biodiversity prediction require additional datasets and trained models.

5) Safety-zone recommendations should not be treated as navigational or emergency guidance without validated real-time marine observations and domain-specific safety systems.

XIII. FUTURE SCOPE

Future work can include real-time ocean and weather APIs, larger marine biodiversity databases, ESP32/Arduino sensor integration, cloud deployment, time-series models, satellite image segmentation, automated anomaly detection, and stronger data validation.

AI models can be selected according to data type: tree-based models for structured environmental variables, temporal models for time-series data, and CNN/U-Net-style approaches for satellite or underwater imagery. The system can also be extended with authentication, downloadable reports, mobile access, multilingual interfaces, and geographically validated safety recommendations.

XIV. CONCLUSION

OceanVerse AI demonstrates a unified software architecture for combining oceanographic monitoring, marine biodiversity information, AI-assisted prediction, satellite analysis, IoT monitoring, interactive mapping, and alerts. The current prototype provides a practical foundation rather than a fully operational marine decision system. Its Random Forest algal-bloom demonstration and Sentinel-2 NDWI workflow show how environmental and Earth-observation data can be brought into one platform.

With larger validated datasets, real-time sensor integration, stronger satellite processing, and domain-specific model validation, OceanVerse AI can be developed into a more robust decision-support platform for marine research, conservation, and safety applications.

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